

Amanda Lucker

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EDUCATION

University of Southern California | Los Angeles, CA

GPA: 3.73/4.00

B.S. Mechanical Engineering

August 2021 – May 2025

Honors: USC Viterbi School of Engineering Dean's List (Fall 2021 - Spring 2025), USC Viterbi Outstanding Ingenuity and Performance

EXPERIENCE

Northrop Grumman

El Segundo, CA

Manufacturing Engineer

September 2025 - Present

- Own deployment, integration, and troubleshooting of 50+ precision laser projection systems, diagnosing hardware, software, and laser accuracy failures to maintain operational uptime for production floor
- Debugged 18 automated fiber placement (AFP) production programs through root cause investigation of supplier software and G-code, executing corrective action plans that restored composite layup and improved machine efficiency by 10%
- Led 6 vendor technical evaluation to replace critical manufacturing equipment, evaluating vendor performance data and coordinating cross-functional stakeholder alignment to deliver a procurement recommendation within a \$500k+ budget
- Define and execute hardware validation test plans, test procedures, and acceptance criteria to qualify manufacturing AFP equipment for post-retrofit production readiness

Manufacturing Engineering Intern

Palmdale, CA | May 2024 - July 2024

- Owned design, fabrication, and test of 5 prototype iterations of a 3D-printed fixture in CATIA, analyzing failure modes to transition an improved design into production, decreasing weekly nuisance failures by 10%
- Redesigned injection molding tooling pneumatics and umbilical cable routing to eliminate interferences and equipment failures

USC Measurement and Instrumentation Laboratory

Los Angeles, CA

Course Producer

August 2024 - May 2025

- Integrated sensors, DAQ hardware, and instrumentation using LabVIEW for hands-on data acquisition, communicating technical concepts clearly to 40 third-year mechatronic students across a multi-disciplinary lab environment
- Applied engineering fundamentals in operating precision measurement instrumentation including oscilloscopes, waveform generators, and Arduino-based circuit assembly for hardware, build, test, and troubleshooting

LEADERSHIP AND INVOLVEMENT

USC FSAE Electric Racing Club

Los Angeles, CA

Powertrain Team

August 2023 - May 2024

- Simulated thermal behavior of a battery cell in ANSYS by applying heat transfer fundamentals, enhancing battery performance
- Designed electro-mechanical accumulator components in Siemens NX, integrating electrical and mechanical requirements while collaborating cross-functionally across all FSAE subteams to proactively resolve design conflicts

Society of Women Engineers, USC Collegiate Section, Member since 2022

Los Angeles, CA

Vice President of Corporate Affairs

April 2024 - May 2025

- Planned and executed 2 professional networking events with 100+ attendees spanning aerospace, biomedical, and CS industries, owning a \$3,000 budget and coordinating logistics to connect 80+ club members with industry professionals

ACADEMIC PROJECTS

Superelastic Tires Best Suited for Space Exploration

July 2024 - May 2025

- Owned hardware design from concept to prototype for an 8-inch scaled NASA Superelastic Tire replica with a \$1,000 budget
- Established an efficient coiling routine to streamline the hand-fabrication of 110 coiled wires, accelerating build time by 80%
- Machined shaft collars on a mill to correct a fit tolerance issue and coordinated with machinist producing lathed aluminum parts
- Defined and executed a thermal test plan using 3 thermocouples, verifying hardware performance across extreme-temperature cycles of shape memory alloy heat treatment to $\pm 2^\circ\text{C}$ accuracy
- Engineered 84 component test bed in SolidWorks with LabView integration to evaluate bead angle impact on tire load capacity
- Chosen out of 39 teams to represent USC Viterbi AME at the AIAA Student Regional Conference, placing 2nd overall

Centrifugal Governor Optimization

March 2025 – May 2025

- Designed and iteratively refined a centrifugal governor mechanism for high-speed rotation using FEA in SolidWorks Simulation
- Built a parametric CAD model with equation-driven dimensions to screen 100+ parameter combinations, reducing simulation run time by 85% and converging efficiently to an optimized final configuration from defined design constraints

SKILLS

Engineering Fundamentals & Design: Structural/Thermal Testing, CAD & Drawings, GD&T, Root Cause Analysis, Environment Testing

Electro-mechanical & Fabrication: Milling, CAD (SolidWorks, CATIA, Siemens NX) MATLAB, ANSYS, FEA, Python, LabVIEW